

INCOME CLASSIFICATION & CUSTOMER SEGMENTATION FOR RETAIL BANKING MARKETING STRATEGY

EXECUTIVE SUMMARY:

This project addresses a retail banking marketing use case using weighted U.S. Census Current Population Survey data (1994-1995) containing 40 demographic and employment-related variables with income labels. This report presents two complementary machine learning solutions for retail banking marketing optimization:

- **Income Classification:** Developed and compared two machine learning models to predict customer income levels ($> \$50K$ vs $< \$50K$) for personalized product offerings.
- **Customer Segmentation:** Identified three distinct customer segments using K-Means clustering for targeted marketing strategies.

The final results show that XGBoost performed best, achieving a strong ROC-AUC (0.9290) and high precision (0.52) with an acceptable recall (0.58) for income classification, while K-Means clustering identified three distinct customer segments to support targeted marketing strategies.

INCOME CLASSIFICATION

DATA INSPECTION:

Dataset shape: 195,615 rows $\times$ 42 columns

Target distribution: 93.8% ($< \$50K$), 6.2% ($\geq \$50K$) - highly imbalanced

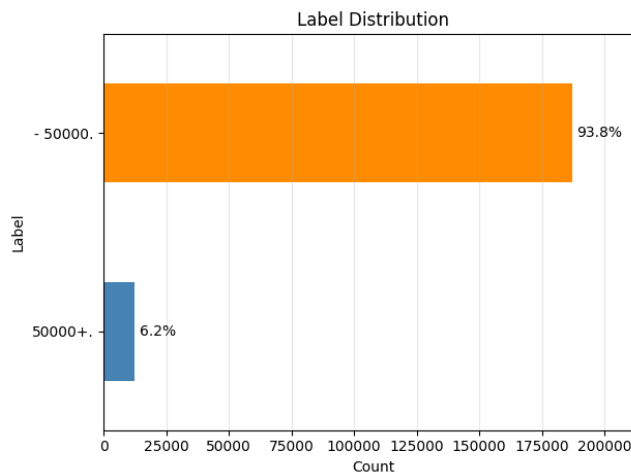


Figure 1: Income Label Distribution Highlighting Majority Low-Income Population

After an initial data audit, 3,908 duplicate records were identified and removed. Features with limited relevance to income prediction or marketing insights, such as parental country of birth and Hispanic origin, were excluded. Variables with substantial missingness (approximately 50% or higher), including migration-related codes, were dropped to avoid noise and unreliable imputation. Additionally, overly granular features, such as detailed industry and occupation recodes, were removed to improve robustness and interpretability.

The final feature set included demographic, education, and employment-related attributes such as age, education level, occupation category, industry pay level, and weeks worked per year, which showed clear relationships with income and were relevant for marketing use cases.

EXPLORATORY DATA ANALYSIS

Numerical Features

Univariate Analysis

Distribution:

To understand how individual features vary across income groups, the distributions of all numerical variables were analyzed and stratified by income level. Based on this analysis, the four most informative features are highlighted below, as they exhibit the strongest and most interpretable differences between individuals earning below \$50,000 and those earning above \$50,000.

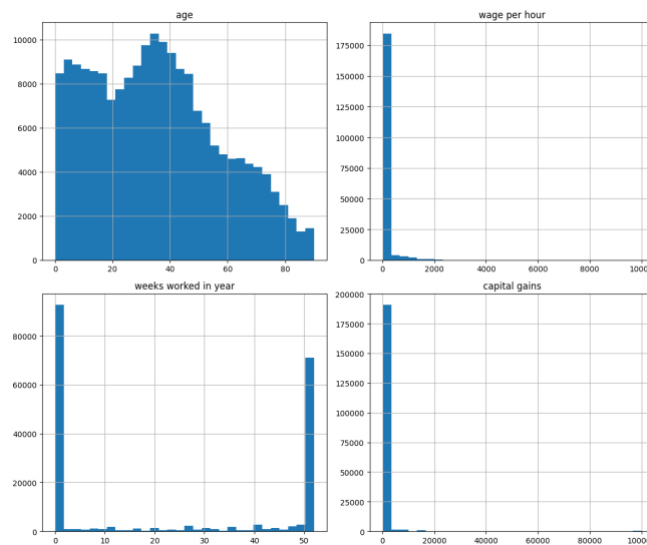


Figure 2: Economic Activity and Income-Related Feature Distributions

Outliers:

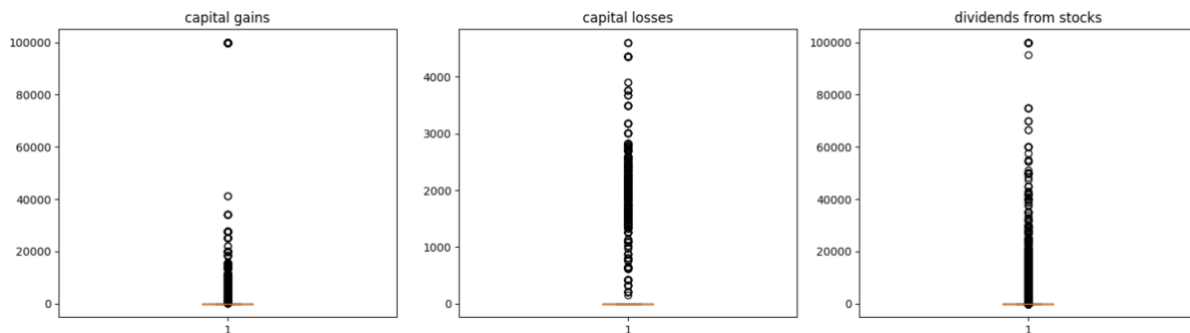


Figure 2: Outlier Distribution of Investment-Related Variables

These variables are highly right-skewed, with most observations at zero and a small number of extreme outliers. This suggests that modeling them as binary indicators (presence vs. absence) is more informative and stable than using raw continuous values.

Bivariate Analysis:

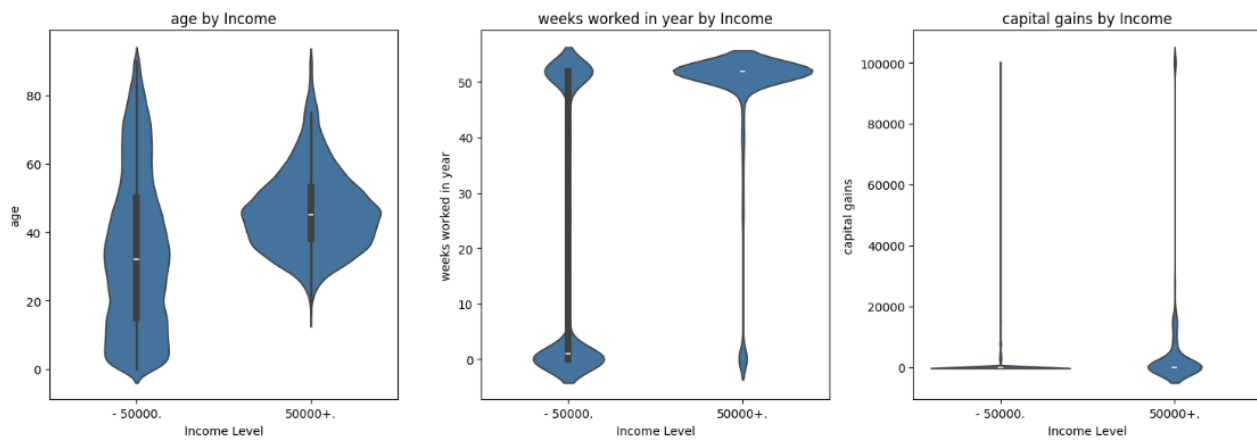


Figure 3: Relationship Between Income Level and Key Demographic and Financial Variables

The violin plots highlight distributional differences rather than just averages, illustrating both central tendency and spread. Weeks worked per year shows the strongest signal, with higher-income individuals concentrated around full-year employment. Capital gains are highly skewed toward the >\$50K group, indicating strong wealth-related effects. Age also differs meaningfully, with higher earners skewing older.

Categorical Features

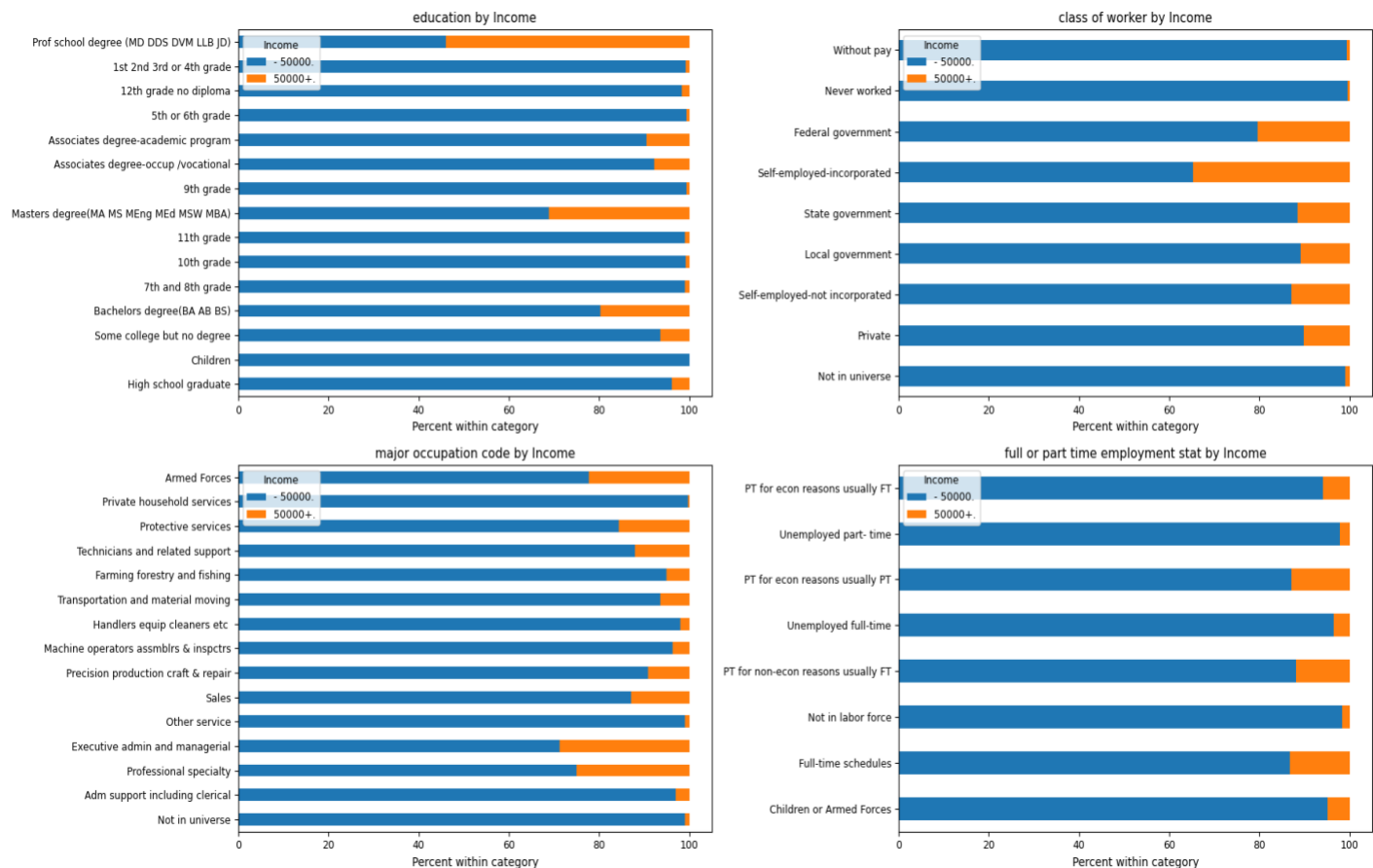


Figure 4: Income Distribution Across Education, Employment Type, Occupation, and Work Status

Analysis of key categorical variables by income shows clear structural differences between groups. Higher-income

individuals are more concentrated in higher education levels, professional and managerial occupations, private or incorporated employment, and full-time work schedules, while lower-income individuals are more prevalent in lower education tiers, service roles, and part-time or non-working statuses.

FEATURE ENGINEERING

To improve model interpretability and reduce feature sparsity, several raw categorical variables were consolidated into higher-level, semantically meaningful categories. The following steps were taken to refine the dataset and improve modeling effectiveness:

- Grouped high-cardinality categorical variables (e.g., education, occupation, industry, household and tax status) into interpretable tiers to reduce sparsity and improve signal quality.
- Converted skewed financial variables (capital gains, losses, dividends) into binary indicators to improve robustness and interpretability.
- Created a full-year employment indicator to capture income stability.
- Removed low-variance and low-coverage features to reduce noise and focus on informative attributes.

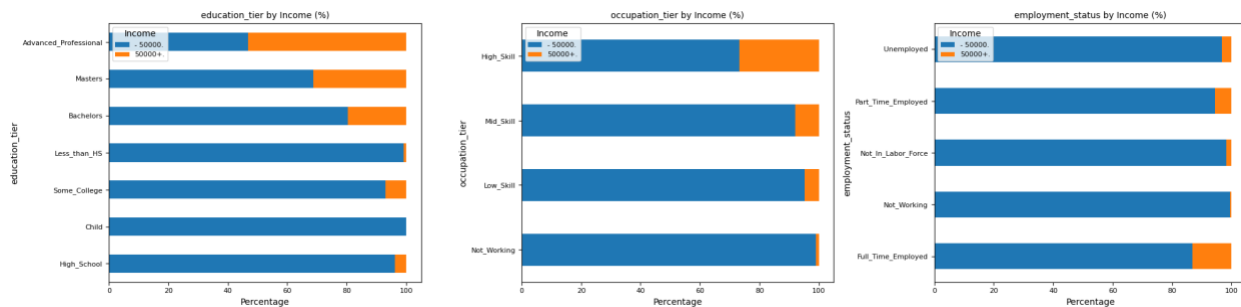


Figure 5: Post-Feature Engineering View of Key Income-Driving Segments

After feature engineering, detailed categorical variables were consolidated into fewer, interpretable tiers, reducing sparsity and clarifying income-related patterns. Education level, occupation skill tier, and employment status all show clear separation, with higher education, high-skill occupations, and full-time employment strongly associated with incomes above \$50K.

MODELING:

Conducted a comparative analysis of logistic regression and XGBoost models across different probability thresholds to evaluate trade-offs between precision and recall for marketing use cases. This analysis helped identify an optimal threshold aligned with targeting efficiency and business objectives.

Logistic Regression:

Logistic Regression was implemented as a baseline classification model using class-balanced weights to handle income imbalance and provide an interpretable benchmark.

Despite its simplicity, the model achieved a strong ROC-AUC of 0.93, indicating high discriminative power between the < \$50K and > \$50K income groups.

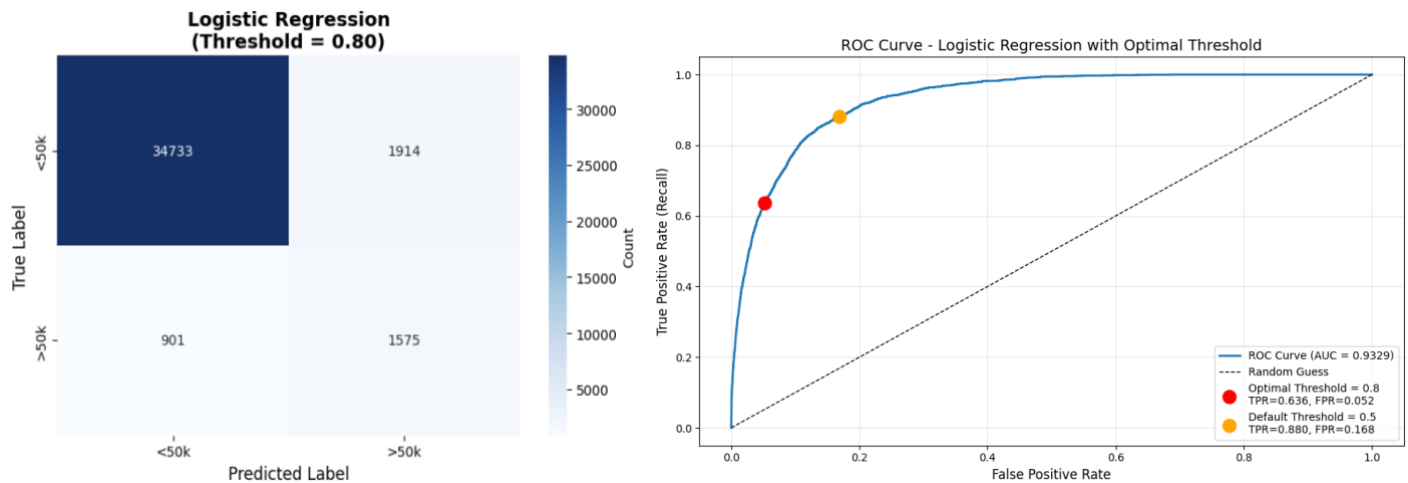


Figure 6: Logistic Regression Performance Using a Precision-Focused Decision Threshold

At its best threshold, for the >50K class, the model achieves a high recall of 0.88 and a precision of 0.26, indicating it captures most high-income individuals while allowing more false positives.

The ROC curve shows that while the model has strong discriminative power ($AUC \approx 0.93$), adjusting the decision threshold from 0.5 to 0.8 significantly reduces false positives at the cost of recall, making it more suitable for marketing use cases where higher precision is preferred meaning a larger proportion of contacted individuals are truly high-income customers.

XGBoost:

XGBoost was applied as a more expressive, non-linear model to improve upon the baseline by capturing complex interactions among demographic and employment features.

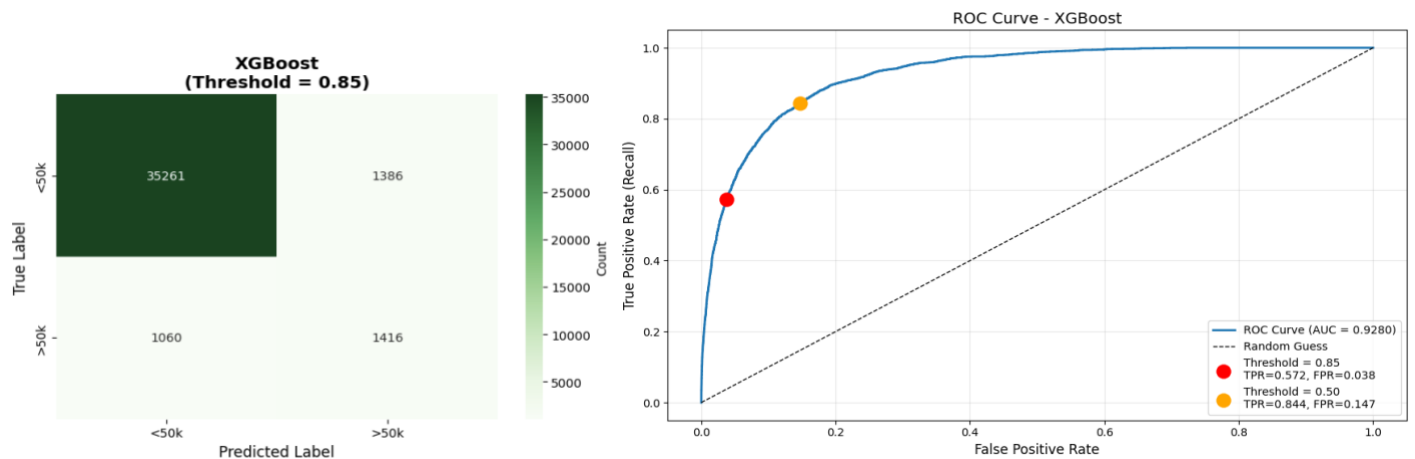


Figure 7: XGBoost Model Performance Using a Precision-Optimized Threshold

The model achieved a strong ROC-AUC of 0.93, with improved precision (0.28) for the >50K class while maintaining high recall (0.84), making it better suited for targeted marketing where accurately identifying high-income individuals is critical.

Model Evaluation:

This evaluation uses 5-fold stratified cross-validation to reliably measure XGBoost's ability to separate income classes while preserving class balance in each fold. The mean CV ROC-AUC of 0.9281 indicates consistently

strong discrimination performance across folds, suggesting the model generalizes well rather than overfitting to a single train-test split.

Model Comparison:

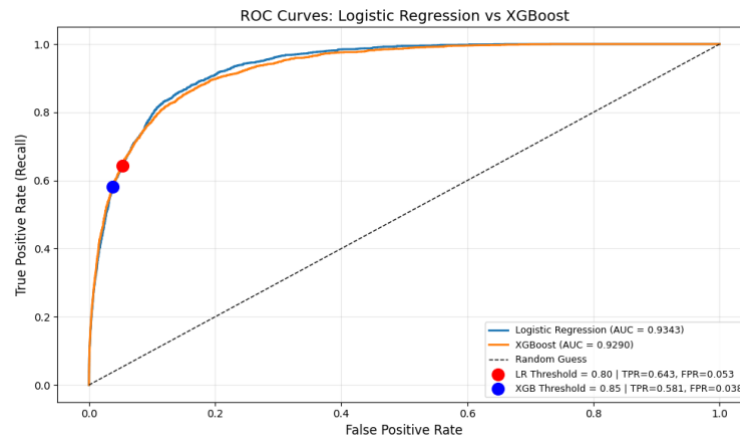


Figure 7: Model Trade-off Between Discrimination Power and False-Positive Control

Metric	Logistic Regression	XGBoost
ROC-AUC	0.9343	0.9290
Precision	0.4650	0.5220
Recall	0.6430	0.5800
F1 Score	0.5400	0.5490

When evaluated at their respective optimal thresholds, Logistic Regression achieves a slightly higher ROC-AUC, indicating a stronger overall ability to distinguish between high-income and low-income individuals.

However, from a marketing perspective, the primary objective is to minimize wasted ad spend on individuals who earn less than \$50K but are incorrectly predicted as high-income. Since these false positives directly translate to unnecessary marketing costs, precision is prioritized over overall discrimination. At its optimal threshold, XGBoost delivers higher precision with fewer false positives, making it the more suitable model for targeted marketing campaigns.

Business Insight: Out of 100 people predicted to earn over \$50K, XGBoost correctly targets about 52 while Logistic Regression targets about 47. XGBoost reduces wasted marketing spend by incorrectly targeting roughly 5 fewer low-income customers per 100 predictions, translating directly into lower outreach costs and more efficient budget utilization.

MODEL INTERPRETABILITY via SHAP Analysis

SHAP analysis was conducted to interpret the model's predictions by quantifying the contribution of each feature to income classification outcomes.

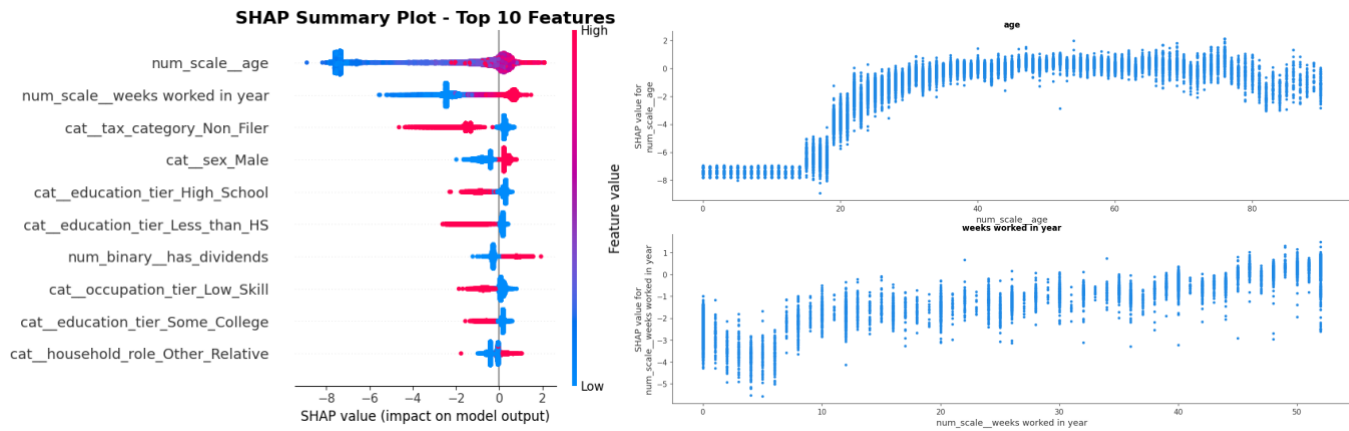


Figure 7: SHAP-Based Feature Importance and Impact on Income Prediction

SHAP analysis shows that age, work stability, education, and capital income drive high-income predictions, with higher age, weeks worked, and dividends increasing the likelihood of earning over \$50K, while low education, non-filing tax status, and low-skill jobs push predictions toward lower income.

Age is the dominant driver, with negative impact below 25 and positive impact from 30- 60, while full-year employment, financial indicators, and skilled occupations increase income predictions, and low-skill occupations consistently predict lower income.

CUSTOMER SEGMENTATION:

For the segmentation task, the focus was on variables that best capture who people are, where they are, and how they participate in the workforce. Based on this, the most suitable features fall into demographic, geographic, and behavioral/economic categories, making them well-aligned for meaningful and actionable customer segments.

Modeling:

For customer segmentation, K-Means clustering was used and evaluated K values from 2 to 7 using both the elbow method (inertia) and silhouette score on a representative random sample of 50,000 observations to ensure computational efficiency.

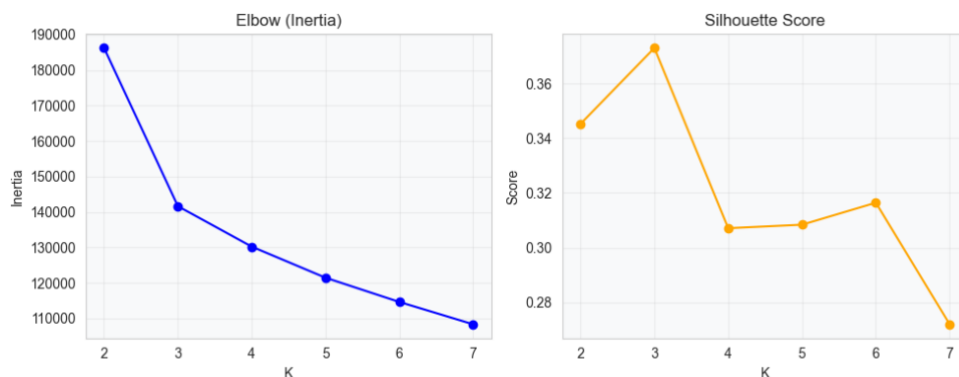


Figure 8: K-Means Cluster Selection Using Elbow and Silhouette Methods

The results indicate that K = 3 is optimal, as it achieves the highest silhouette score while showing a clear slowdown in inertia reduction, suggesting well-separated and meaningful customer segments.

Segment Interpretation:

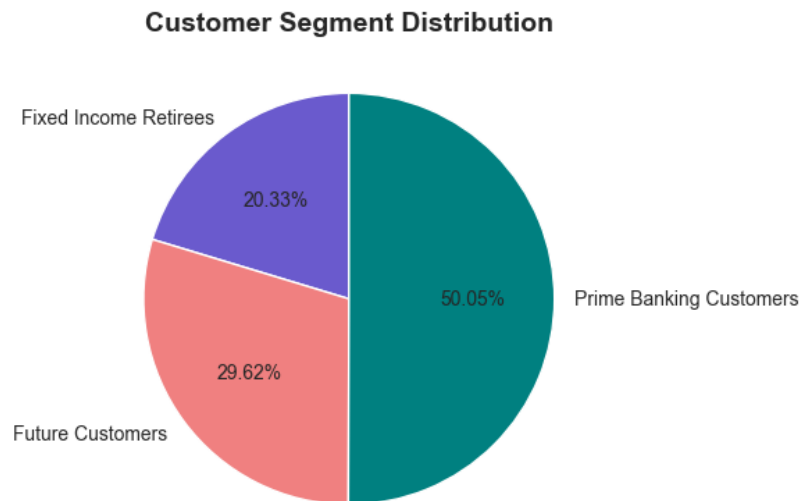


Figure 9: Customer Segment Distribution Based on K-Means Clustering

The segmentation reveals a well-balanced customer base with clear strategic value: Prime Banking Customers (~50%) represent the core revenue-driving segment suited for premium products and cross-selling, Future Customers (~30%) show strong growth potential and can be nurtured through targeted engagement, and Fixed Income Retirees (~20%) form a stable, lower-risk group best served with conservative offerings. These segment names were chosen to reflect both current financial behavior and long-term business value, and the rationale behind this naming will be explained later in the analysis.

Demographic Profile

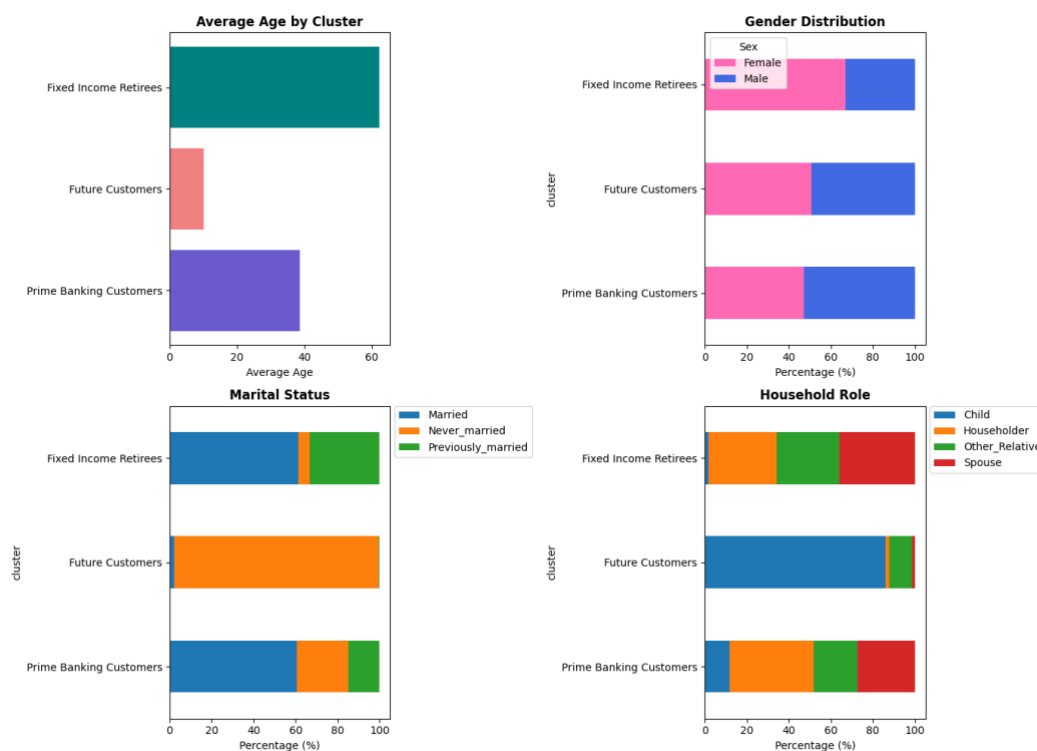


Figure 9: Demographic Profile of Customer Segments

The clusters show clear life-stage differences where Prime Banking Customers are middle-aged, mostly married householders, while Future Customers are very young, predominantly never-married dependents. Fixed Income Retirees are the oldest group, largely married or previously married, with household roles centered around spouses or relatives which makes each segment distinct for targeted marketing.

Socioeconomic Profile

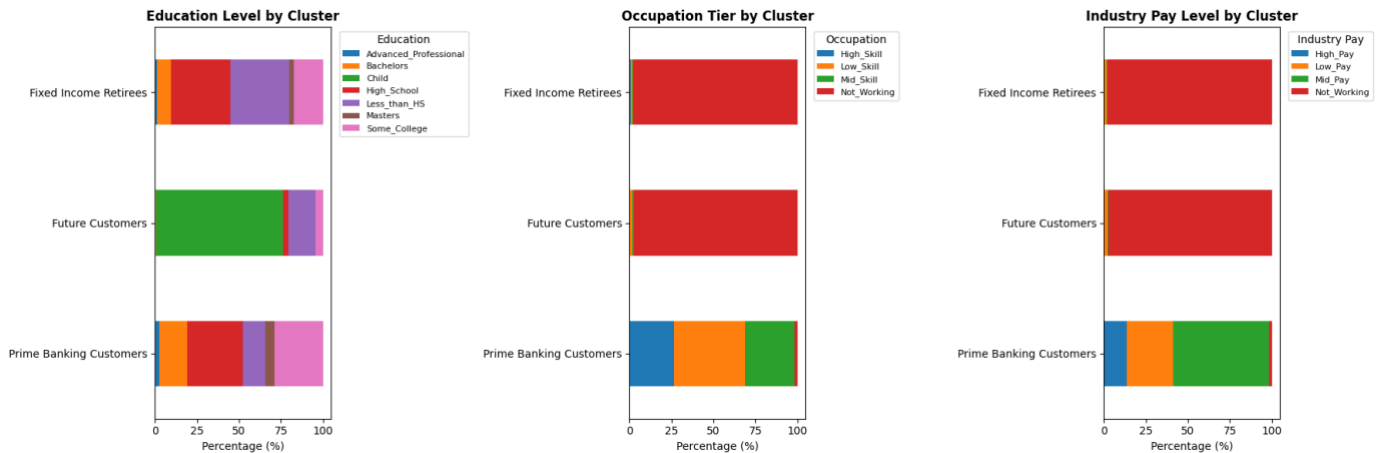


Figure 10: Socioeconomic Characteristics of Customer Segments

Prime Banking Customers are highly educated and concentrated in higher-skill, higher-paying roles, while Future Customers have moderate education but lower workforce participation. Fixed Income Retirees are largely not working, reflecting stable, retirement-based income. This enables premium targeting for Prime Banking Customers, growth-focused strategies for Future Customers, and conservative offerings for Fixed Income Retirees.

Employment Pattern



Figure 11: Workforce Participation Differences by Customer Segment

Prime Banking Customers show strong labor attachment, with a high share of full-year workers, the highest average weeks worked, and a mix of private, government, and self-employed roles. Future Customers and Fixed Income Retirees exhibit minimal full-year employment and very low average weeks worked, indicating limited or no current labor income. From a business perspective, this reinforces prioritizing premium, income-linked financial products for Prime Banking Customers, while focusing on long-term growth pathways for Future Customers and stable, low-risk offerings for Fixed Income Retirees.

Marketing Perspective:

Prime Banking Customers (Primary Target): High-earning, actively employed customers with the highest transaction volume. They can be marketed through digital channels with credit, lending, and investment products focused on convenience and flexibility.

Fixed Income Retirees (Steady Revenue): Asset-rich and highly loyal customers who value trust and personal service, making retirement planning, savings products, and branch-based engagement most effective. They can be marketed through phone with saving accounts and retirement planning.

Future Customers (Long-Term Investment): Younger, low-current-value customers with strong lifetime potential. They can be marketed through social media or school-focused programs with youth accounts.

Budget Allocation:

At the outset, marketing budget is allocated across the three segments based on their relative priority scores. The raw scores (75-14-11) are calculated for Prime Banking Customers, Fixed Income Retirees and Future Customers respectively and adjusted to strategic allocation (60-25-15) because Fixed Income's asset-based value isn't captured by activity metrics, and Prime Banking faces diminishing returns beyond 60%

$$\text{Priority Score} = 0.30 * \text{Market Size} + 0.50 * \text{Weeks Worked} + 0.20 * \text{Income Potential}$$

The weights prioritize near-term ROI and execution efficiency. Weeks worked (50%) captures the number of weeks a person worked in a year, market size (30%) reflects scale and reach, and income potential (20%) accounts for upside value without over-relying on theoretical wealth.

Recommended Marketing Budget Allocation

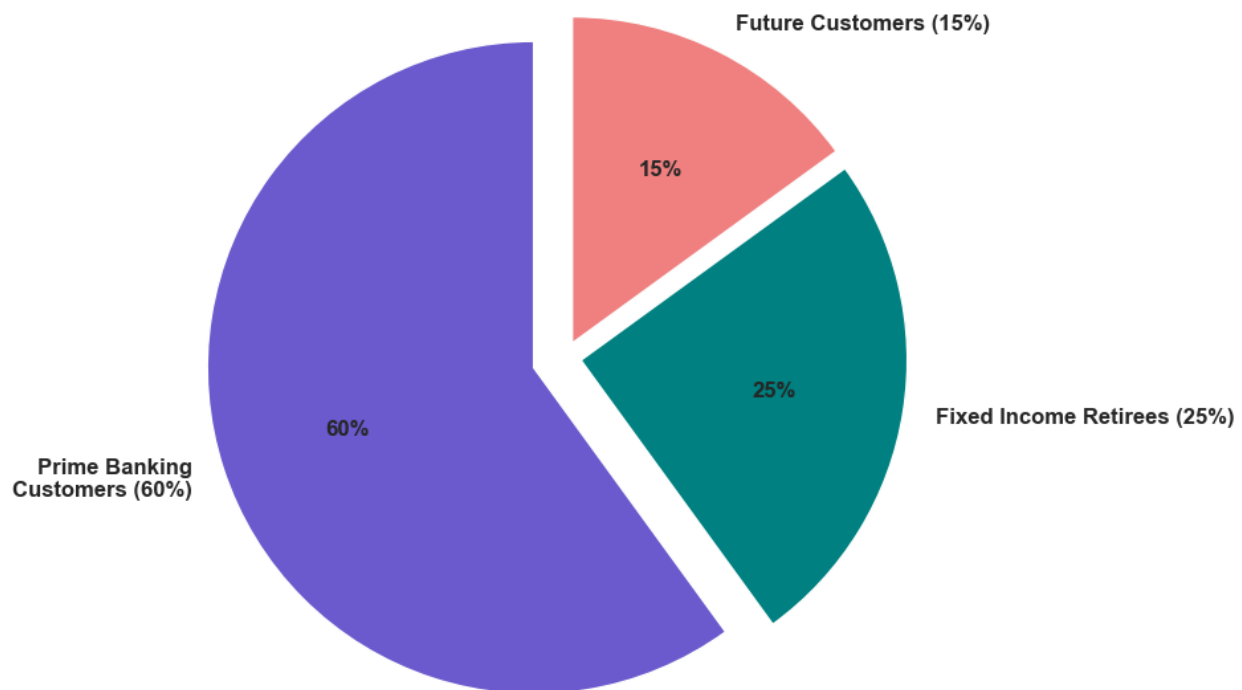


Figure 12: Strategic Allocation of Marketing Spend Across Customer Segments

References:

- <https://www.aidancooper.co.uk/a-non-technical-guide-to-interpreting-shap-analyses/>
- <https://www.analyticsvidhya.com/blog/2021/06/how-to-solve-customer-segmentation-problem-with-machine-learning/>
- <https://openviewpartners.com/blog/customer-segmentation/>
- <https://improvado.io/blog/marketing-budget-allocation>
- <https://visbanking.com/customer-segmentation-in-banking>